

host_agent

Here is a summary of everything Ordnance Survey (OS) knows about electricity in Exeter:

- Land use features related to electricity in Exeter include:
- 558 Electricity Sub Stations
- 6 Electricity Distribution Sites
- 2 Power Stations

Building features related to electricity in Exeter include:

- 355 Electricity Sub Stations
- 4 Electricity Distribution Facilities
- 2 Power Stations

All these features are mapped within the Exeter area boundary. The map includes color-coded land use and building types, and you can explore the locations and details interactively.

If you would like to view or download the interactive map, let me know! If you need further breakdowns or details about specific features, I can provide those as well.

planning_agent

The query is focused on gathering comprehensive information about electricity in Exeter. The geographical area is Exeter, which is one entry. The entities to be searched are related to electricity, which may include infrastructure, services, and statistics. Since the user wants to see everything, we will need to consider various types of artifacts that Ordnance Survey may have, such as maps, datasets, and reports.

output steps: 1. Find Exeter 1 area 2. Search for Ordnance Survey datasets related to electricity in Exeter (including infrastructure, services, and statistics) 3. Search for Ordnance Survey maps that depict electricity infrastructure in Exeter 4. Search for any reports or publications by Ordnance Survey that discuss electricity in Exeter 5. Compile and present all findings related to electricity in Exeter from the above searches.

named_area_agent

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) specifically for Exeter.

data_analysis_agent

Filtered polygons for Exeter have been found. There are 34 matching entries. The results are saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) specifically for Exeter, filtered by name, description, or settlement group.

No electricity-related structures such as substations, transformers, or power stations were found in the provided dataset. The dataset does not contain any entries matching these keywords in the relevant columns. If you need further analysis or wish to search for other types of electricity infrastructure, please specify.

structures_agent

No specific electricity-related structures such as substations, transformers, or power stations were found in the dataset for the 'filtered_exeter_area_polygon'. The search included structures like electricity pylons and masts, but none matched the additional criteria. If you need further assistance or wish to explore other types of structures, please let me know!

land_use_features_agent

I found 566 records related to electricity features within the 'filtered_exeter_area_polygon'. The data is saved in the artifact named 'landuse_electricity_features_exeter'.

If you need further analysis or filtering, please let me know!

buildings_agent

I found buildings related to electricity within the 'filtered_exeter_area_polygon' artifact. The search returned 72,895 records. The data is saved in the artifact named `buildings_electricity_buildings_exeter`.

plotting_agent

Exeter electricity-related features: - Landuse features: {'Electricity Sub Station': 558, 'Electricity Distribution Site': 6, 'Power Station': 2} - Building features: {'Electricity Sub Station': 355, 'Electricity Distribution Facility': 4, 'Power Station': 2}

First 5 landuse features: 1. Electricity Distribution Site (various polygons and multipolygons) 2. Electricity Distribution Site 3. Electricity Distribution Site 4. Electricity Distribution Site 5. Electricity Distribution Site

First 5 building features: 1. Electricity Distribution Facility (polygon) 2. Electricity Distribution Facility 3. Electricity Distribution Facility 4. Electricity Distribution Facility 5. Electricity Sub Station

An interactive map has been generated showing all OS electricity-related landuse and building features in Exeter, with the area boundary. Landuse and building types are color-coded and summarized.

Artifact name: `exeter_electricity_features_map`

Artifact description: A folium map showing all OS electricity-related landuse and building features in Exeter, with area boundary. Landuse and building types are color-coded and summarized.

Artifact data: `exeter_electricity_features_map.html`

Tool Calls and Code

Tool: `send_message`

- Args: {'target': 'planning_agent', 'task_description': 'The user wants to see everything Ordnance Survey knows about electricity in Exeter. Provide a sequence of steps to solve this query, including which agents and artifact types to use for a comprehensive search.'}
- Result: The query is focused on gathering comprehensive information about electricity in Exeter. The geographical area is Exeter, which is one entry. The entities to be searched are related to electricity, which may include infrastructure, services, and statistics. Since the user wants to see everything, we will need to consider various types of artifacts that Ordnance Survey may have, such as maps, datasets, and reports.

output steps: 1. Find Exeter 1 area 2. Search for Ordnance Survey datasets related to electricity in Exeter (including infrastructure, services, and statistics) 3. Search for Ordnance Survey maps that depict electricity infrastructure in Exeter 4. Search for any reports or publications by Ordnance Survey that discuss electricity in Exeter 5. Compile and present all findings related to electricity in Exeter from the above searches.

Tool: send_message

- Args: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}
- Result: The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) specifically for Exeter.

Tool: call_os_ngd

- Args: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}
- Result: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..., count of records fetched : 98.

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}
- Result: Filtered polygons for Exeter have been found. There are 34 matching entries. The results are saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) specifically for Exeter, filtered by name, description, or settlement group.

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['exeter_area_polygon']}
- Result: (metadata generated for 'exeter_area_polygon')

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):  
    gdf = data[0]  
    # Filter for polygons where the name or description is related to 'Exeter'  
    filtered = gdf[(gdf['name1_text'].str.contains('Exeter', case=False, na=False)) |
```

```

        (gdf['description'].str.contains('Exeter', case=False, na=False)) |
        (gdf['descriptiongroup'].str.contains('Settlement', case=False, na=False)) |
        (gdf['name1_text'].str.lower() == 'exeter')]
    return [
        f"Filtered polygons for Exeter. Number of matching entries: {len(filtered)}.",
        'filtered_exeter_area_polygon',
        'GeoDataFrame containing only the polygon(s) specifically for Exeter, filtered by name, description, or settlement group.',
        filtered
    ]

```

Tool: send_message

- Args: {'target': 'structures_agent', 'task_description': "Search for electricity-related structures (such as substations, pylons, transformers, power stations, masts, etc.) within the 'filtered_exeter_area_polygon' artifact."}
- Result: No specific electricity-related structures such as substations, transformers, or power stations were found in the dataset for the 'filtered_exeter_area_polygon'. The search included structures like electricity pylons and masts, but none matched the additional criteria. If you need further assistance or wish to explore other types of structures, please let me know!

Tool: call_os_ngd

- Args: {'filters': ['Electricity Metal Monopole', 'Electricity Pylon', 'Mast'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'electricity_structures_exeter'}
- Result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['electricity_structures_exeter'], Descriptions are : ["A geopandas dataframe containing structure data with filters and bbox applied for the query Search for electricity-related structures (such as substations, pylons, transformers, power stations, masts, etc.) within the 'filtered_exeter_area_polygon' artifact. using filters ['Electricity Metal Monopole', 'Electricity Pylon', 'Mast']. Further Name filtering is available for this"], counts of records fetched are : [22].

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Please analyze the 'electricity_structures_exeter' artifact to filter and provide insights specifically on electricity-related structures such as substations, transformers, and power stations. Save the filtered results as 'filtered_electricity_structures_exeter'."}
- Result: No electricity-related structures such as substations, transformers, or power stations were found in the provided dataset. The dataset does not contain any entries matching these keywords in the relevant columns. If you need further analysis or wish to search for other types of electricity infrastructure, please specify.

Tool: call_os_ngd

- Args: {'filters': ['Electricity Distribution Site', 'Electricity Storage Site', 'Electricity Sub Station', 'Power Station'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'electricity_features_exeter'}
- Result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['landuse_electricity_features_exeter'], Descriptions are : ["A geopandas dataframe containing land use data with filters and bbox applied for the query Search for land use features related to electricity (such as electricity substations, power stations, or energy facilities) within the 'filtered_exeter_area_polygon' artifact. using filters ['Electricity Distribution Site', 'Electricity Storage Site', 'Electricity Sub Station', 'Power Station']. Further Name filtering is available for this"], counts of records fetched are : [566].

Tool: send_message

- Args: {'target': 'buildings', 'task_description': "Search for buildings related to electricity (such as electricity offices, power stations, or substations) within the 'filtered_exeter_area_polygon' artifact."}
- Result: I found buildings related to electricity within the 'filtered_exeter_area_polygon' artifact. The search returned 72,895 records. The data is saved in the artifact named `buildings_electricity_buildings_exeter`.

Tool: call_os_ngd

- Args: {'filters': ['Electricity Distribution Facility', 'Electricity Storage Facility', 'Electricity Sub Station', 'Power Station'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'electricity_buildings_exeter'}
- Result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_electricity_buildings_exeter'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for buildings related to electricity (such as electricity offices, power stations, or substations) within the 'filtered_exeter_area_polygon' artifact. using filters ['Electricity Distribution Facility', 'Electricity Storage Facility', 'Electricity Sub Station', 'Power Station']."], counts of records fetched are : [72895].

Tool: send_message

- Args: {'target': 'plotting_agent', 'task_description': "Plot the following artifacts to show everything OS knows about electricity in Exeter: 'filtered_exeter_area_polygon', 'landuse_electricity_features_exeter', and 'buildings_electricity_buildings_exeter'. Provide a summary of the types of features found."}

- Result: Exeter electricity-related features:
- Landuse features: {'Electricity Sub Station': 558, 'Electricity Distribution Site': 6, 'Power Station': 2}
- Building features: {'Electricity Sub Station': 355, 'Electricity Distribution Facility': 4, 'Power Station': 2}

First 5 landuse features: 1. Electricity Distribution Site (various polygons and multipolygons) 2. Electricity Distribution Site 3. Electricity Distribution Site 4. Electricity Distribution Site 5. Electricity Distribution Site

First 5 building features: 1. Electricity Distribution Facility (polygon) 2. Electricity Distribution Facility 3. Electricity Distribution Facility 4. Electricity Distribution Facility 5. Electricity Sub Station

An interactive map has been generated showing all OS electricity-related landuse and building features in Exeter, with the area boundary. Landuse and building types are color-coded and summarized.

Artifact name: exeter_electricity_features_map

Artifact description: A folium map showing all OS electricity-related landuse and building features in Exeter, with area boundary. Landuse and building types are color-coded and summarized.

Artifact data: exeter_electricity_features_map.html

Tool Code Executor

```
def plot_electricity_features_exeter(data: list):
    import geopandas as gpd
    import folium
    import pandas as pd
    # Unpack data
    exeter_area = data[0]
    landuse_electricity = data[1]
    buildings_electricity = data[2]
    # Ensure all are in EPSG:4326
    exeter_area = exeter_area.to_crs(epsg=4326)
    landuse_electricity = landuse_electricity.to_crs(epsg=4326)
    buildings_electricity = buildings_electricity.to_crs(epsg=4326)
    # Remove all columns except geometry and description for folium
    landuse_electricity = landuse_electricity[['geometry', 'description']].copy()
    buildings_electricity = buildings_electricity[['geometry', 'description']].copy()
    exeter_area = exeter_area[['geometry']].copy()
    # Create base map centered on Exeter
    centroid = exeter_area.geometry.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=12, tiles='cartodbpositron')
    # Plot Exeter area boundary
```

```

folium.GeoJson(
    exeter_area,
    name='Exeter Area',
    style_function=lambda x: {'fillColor': '#00000000', 'color': 'black', 'weight': 2, 'fillOpacity': 0.05}
).add_to(m)
# Plot landuse electricity features
landuse_colors = {
    'Electricity Distribution Site': 'orange',
    'Electricity Sub Station': 'purple',
    'Power Station': 'red'
}
for desc, color in landuse_colors.items():
    subset = landuse_electricity[landuse_electricity['description'] == desc]
    if not subset.empty:
        folium.GeoJson(
            subset,
            name=f'Landuse: {desc}',
            style_function=lambda x, col=color: {'fillColor': col, 'color': col, 'weight': 1, 'fillOpacity': 0.4},
            tooltip=folium.GeoJsonTooltip(fields=['description'])
        ).add_to(m)
# Plot buildings electricity features
building_colors = {
    'Electricity Distribution Facility': 'blue',
    'Electricity Sub Station': 'green',
    'Power Station': 'red'
}
for desc, color in building_colors.items():
    subset = buildings_electricity[buildings_electricity['description'] == desc]
    if not subset.empty:
        folium.GeoJson(
            subset,
            name=f'Building: {desc}',
            style_function=lambda x, col=color: {'fillColor': col, 'color': col, 'weight': 1, 'fillOpacity': 0.6},
            tooltip=folium.GeoJsonTooltip(fields=['description'])
        ).add_to(m)
folium.LayerControl().add_to(m)
# Summary of types of features
landuse_types = landuse_electricity['description'].value_counts().to_dict()
building_types = buildings_electricity['description'].value_counts().to_dict()
summary = (
    f"Exeter electricity-related features:\n"
    f"- Landuse features: {landuse_types}\n"
    f"- Building features: {building_types}\n"
    f"First 5 landuse features:\n{landuse_electricity.head(5).to_string(index=False)}\n"
    f"First 5 building features:\n{buildings_electricity.head(5).to_string(index=False)}"
)
# Save map
map_filename = 'exeter_electricity_features_map.html'

```



```
m.save(map_filename)
```

```
return [
```

```
    summary,
```

```
    'exeter_electricity_features_map',
```

```
    'A folium map showing all OS electricity-related landuse and building features in Exeter, with area boundary. Landuse and building types are color-coded',
```

```
    map_filename
```

```
]
```